A1.2 Nucleic acids

A1 Unity and diversity
Molecules

HL

Understandings:

- A1.2.11—Directionality of RNA and DNA
- A1.2.12—Purine-to-pyrimidine bonding as a component of DNA helix stability
- A1.2.13—Structure of a nucleosome
- A1.2.14—Evidence from the Hershey–Chase experiment for DNA as the genetic material
- A1.2.15—Chargaff's data on the relative amounts of pyrimidine and purine bases across diverse life forms

Key words:

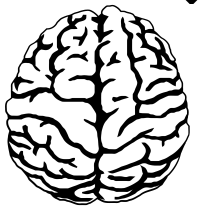
Purine
Pyrimidine
Nucleosome
Histone
Hershey-Chase experiment
Bacteriophage
Supernatant
Sediment
Chargaff
Induction
Tetranucleotide hypothesis

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



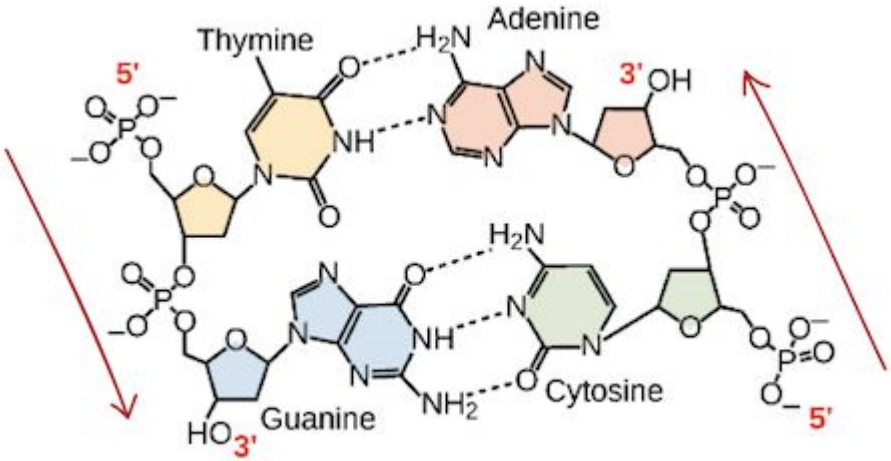
Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



A1.2.11—Directionality of RNA and DNA

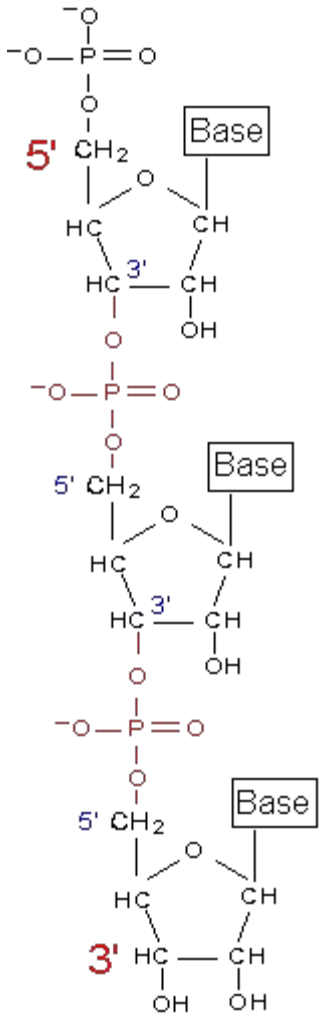
DNA is antiparallel.

While one side goes in a 3’ to 5’ direction, the opposite side goes in the 5’ to 3’ direction



Nucleotides are always added to the 3’ carbon

DNA always grows in the 5’ to 3’ direction



This becomes important in understanding

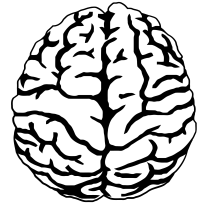
Replication

Transcription

Translation

(more later!)

A1.2.12—Purine-to-pyrimidine bonding as a component of DNA helix stability



The complementary bases of a double strand of DNA fit perfectly together.

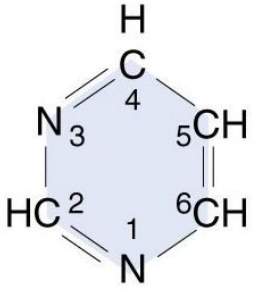
Which bases are always paired together?

The complementary bases of a double strand of DNA fit perfectly together.

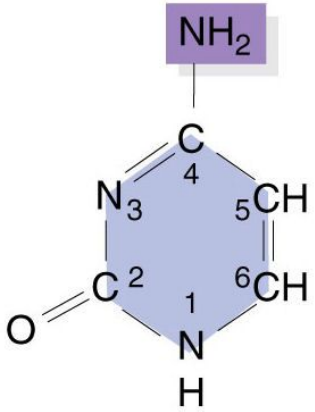
Which bases are always paired together?

Adenine- Thymine

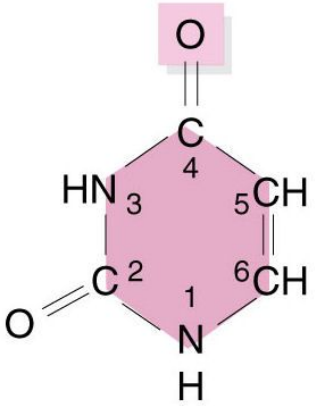
Cytosine-Guanine



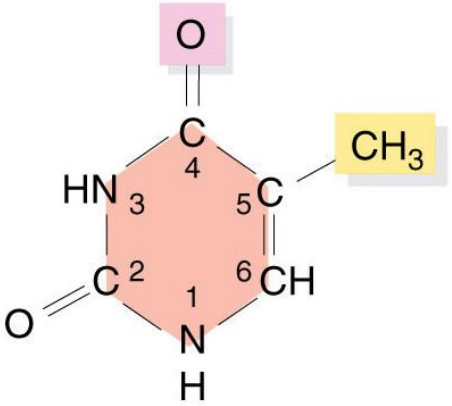
Pyrimidine



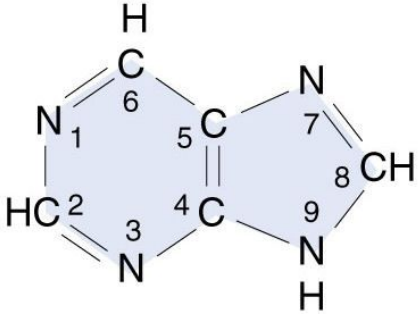
Cytosine (C)



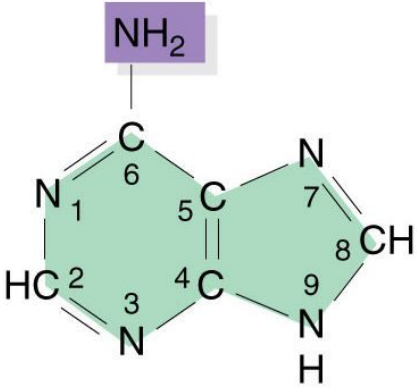
Uracil (U)
(found in RNA)



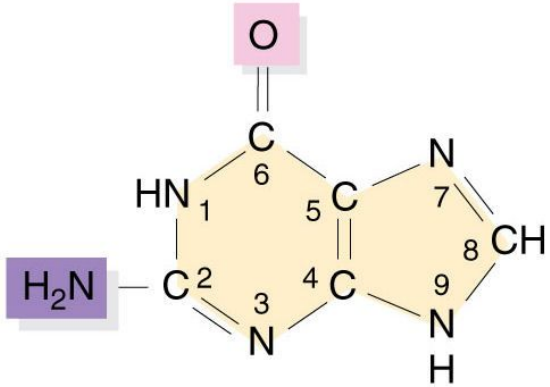
Thymine (T)
(found in DNA)



Purine

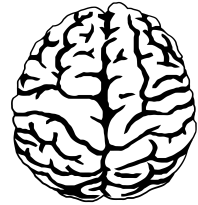


Adenine (A)



Guanine (G)

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Purines always bond to **pyrimidines**

A purine plus a pyrimidine always has the same length - this allows the DNA double helix to have a uniform diameter (2 nanometers)

What type of bond holds purines and pyrimidines together?

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A-T = two hydrogen bonds, C-G = 3 hydrogen bonds

Mnemonic

Guardian **A**ngels are **P**ure

They have two wings!

Guanine, **A**denine, **P**urines

Two rings!



Mnemonic

Pyramids are sharp so they

C U T

Pyrimidines are

Cytosine, **U**racil, **T**hymine

One pyramid, one ring



A1.2.13—Structure of a nucleosome



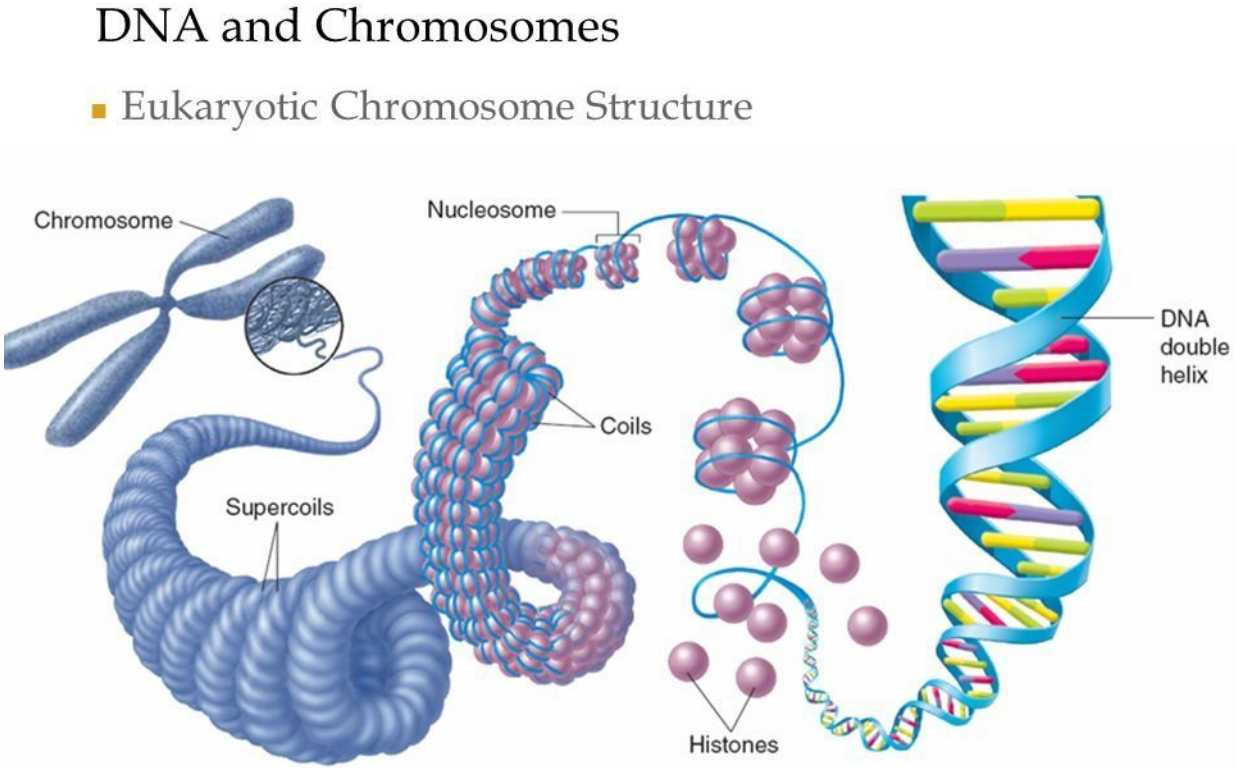
If the DNA within a single human cell was straightened and laid end to end, how long would it stretch?

a) 2cm
b) 2m
c) 2km

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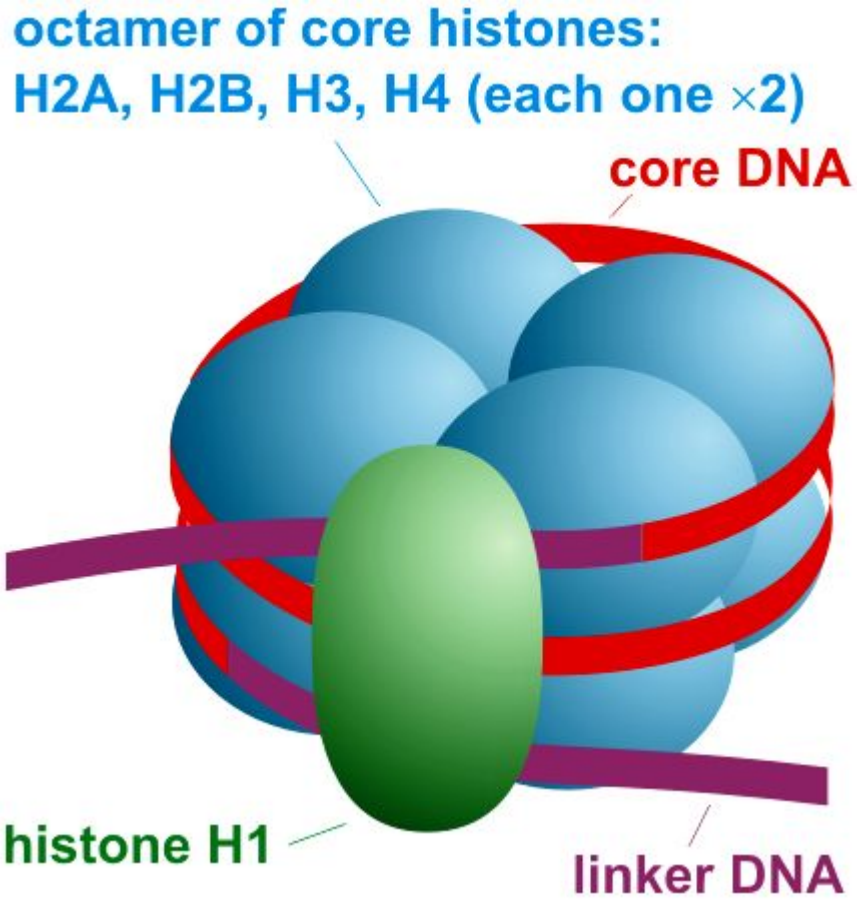
a) 2cm
b) 2m
c) 2km

Nucleosomes help to supercoil the DNA.



Nucleosomes

In order to pack the long DNA found in eukaryotic cells, the structure needs to repeatedly coiled, making it more compact.

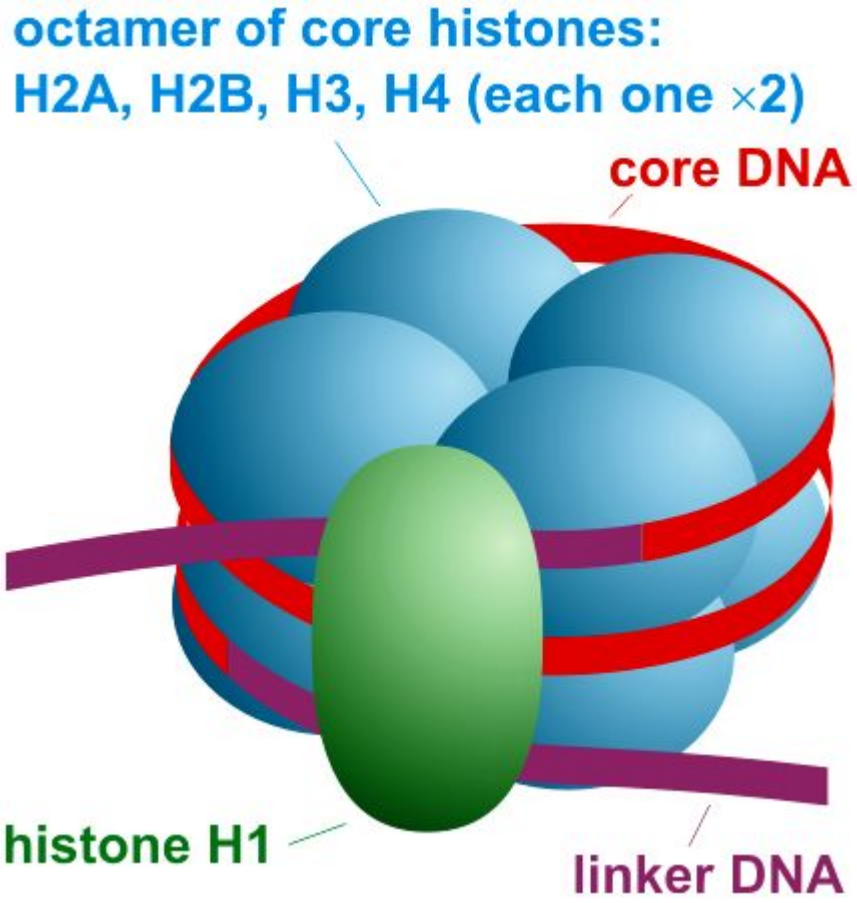


Nucleosomes

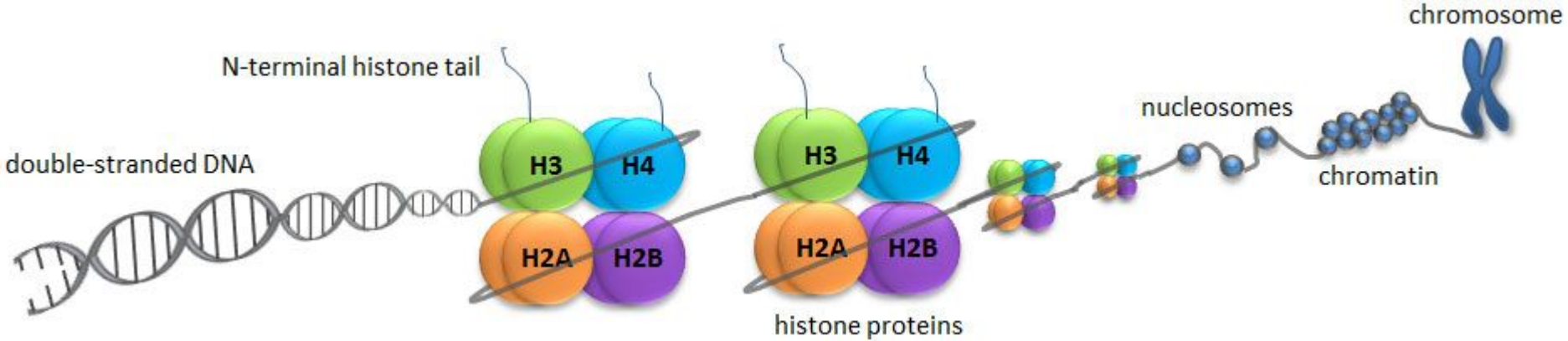
There is a core of 8 histones and a linker histone, H1.

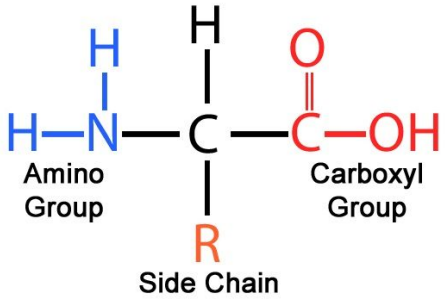
Histones are proteins that adhere to DNA

They bond due to negative charges on the DNA and positive charges on the histone protein



Positively charged amino acids attract negatively charged DNA





Utilization of molecular visualization software to analyse the association between protein and DNA within a nucleosome.

https://www.rcsb.org/

Search “nucleosome”

Select Homo sapiens



A1.2.14—Evidence from the Hershey–Chase experiment for DNA as the genetic material

Hershey-Chase experiment

Since 1800s scientists thought that chromosomes were the material that was inherited.

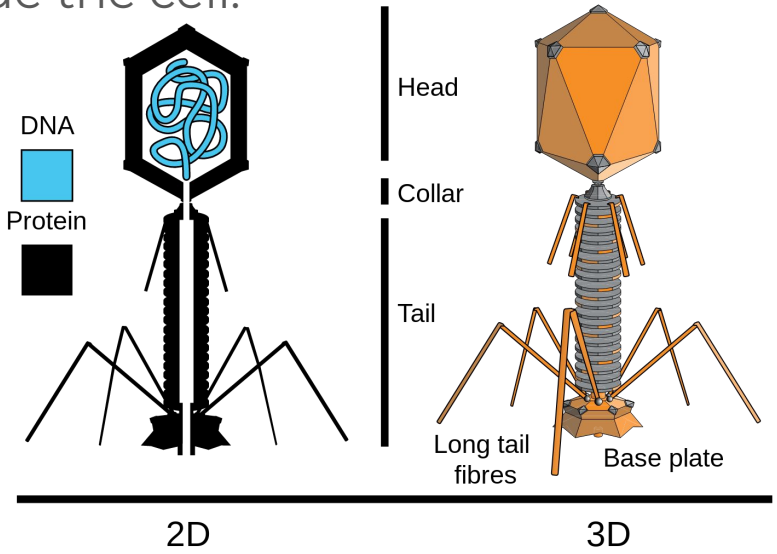
Unsure whether proteins or nucleic acids were the “genetic material”

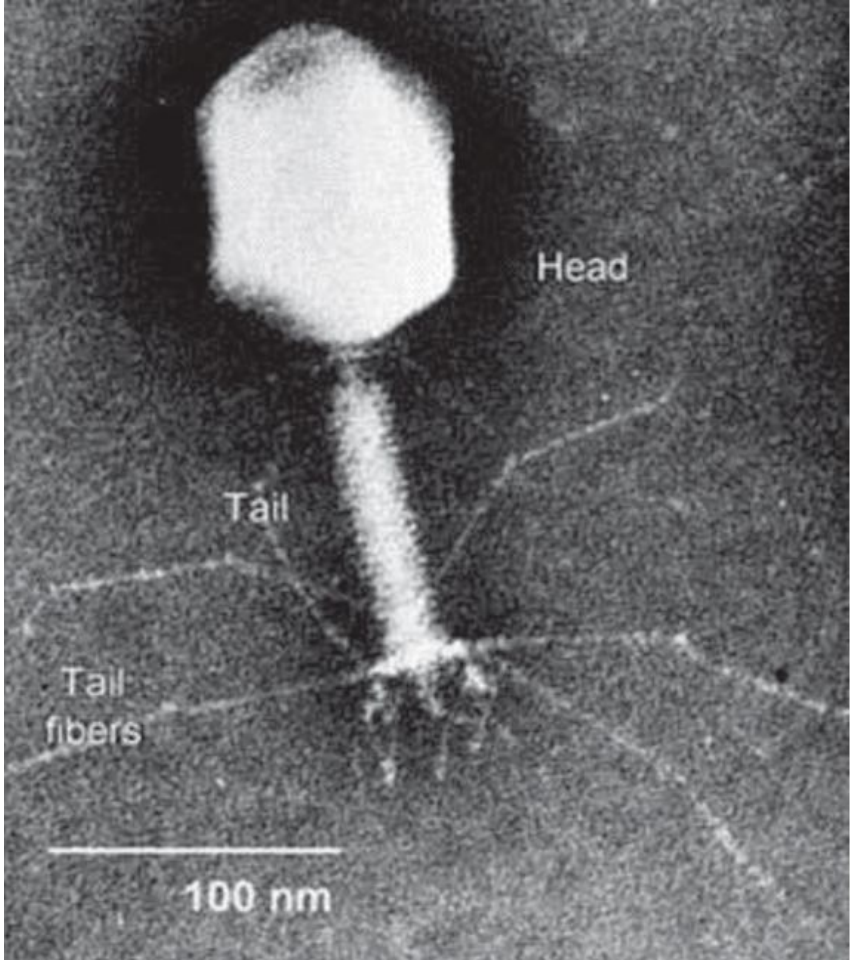
1952 Alfred Hershey and Martha Chase used an experiment with viruses to show that nucleic acid was the genetic material, not protein.

T2 Bacteriophage

Virus that infects bacteria.

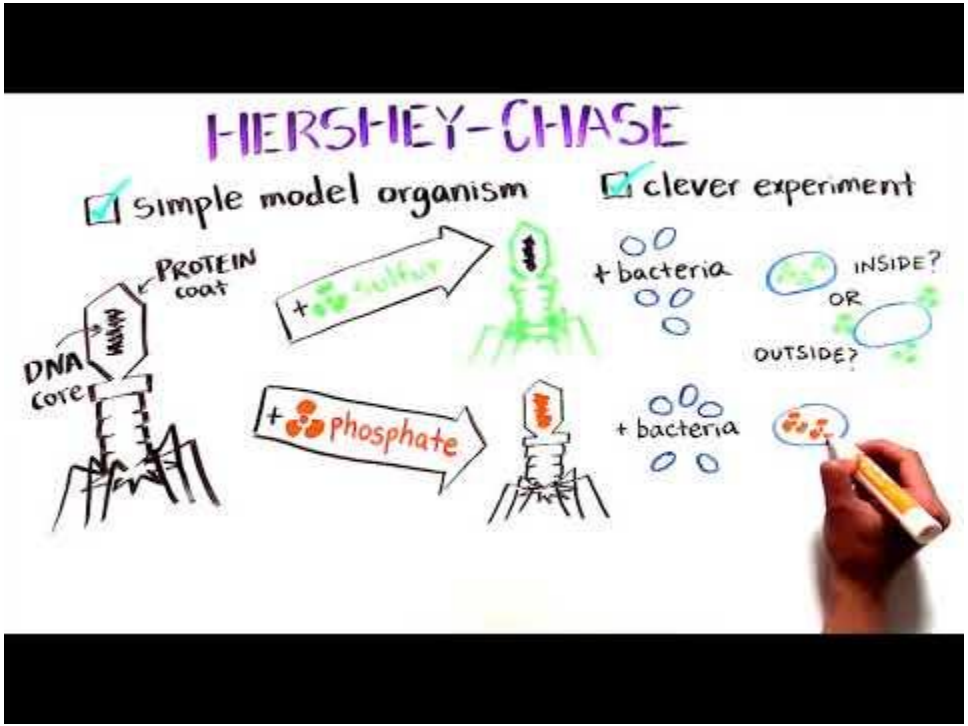
Injects nucleic acid into cell while non genetic portion remains outside the cell.





What type of image do you think this is?

Hershey and Chase



Radioactively label protein (using radioactive sulfur) and DNA (using radioactive phosphate) of bacteriophage

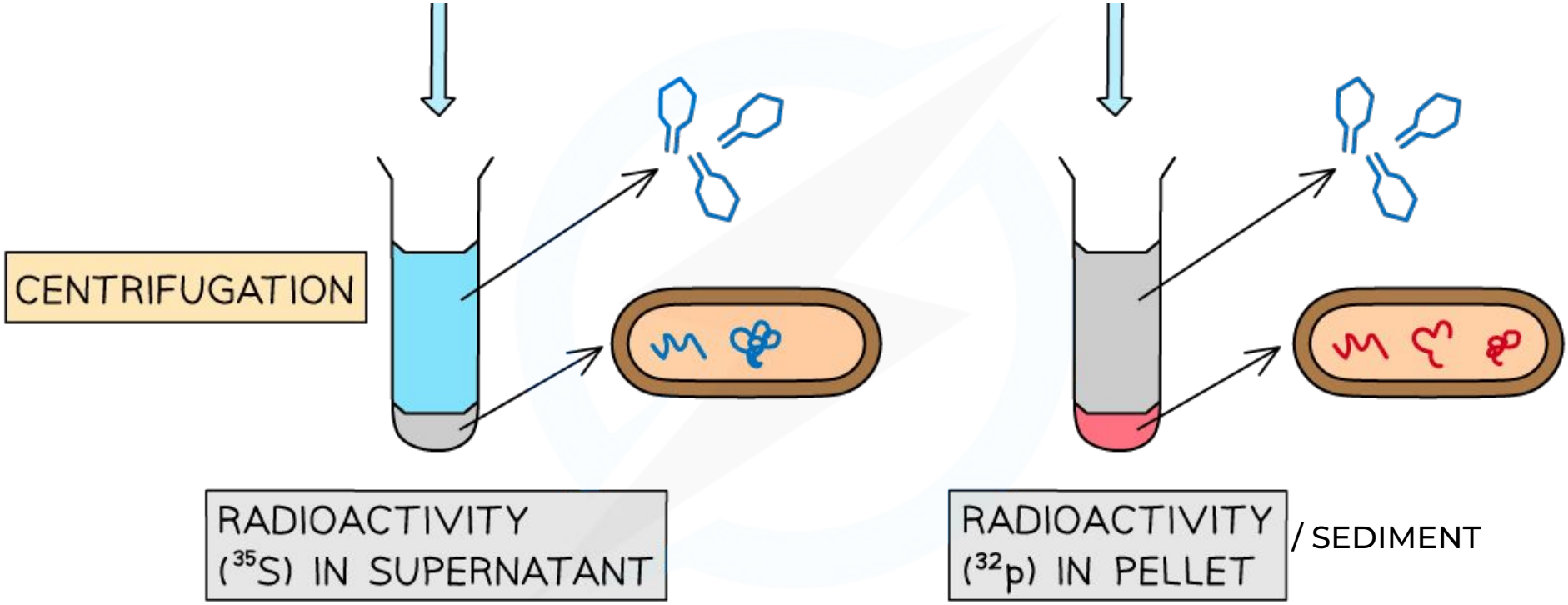
Allow to infect bacterial cells

Waring blender to shake viruses off bacteria

Centrifuge to separate bacterial cells and viruses

Check where radioactive substance is

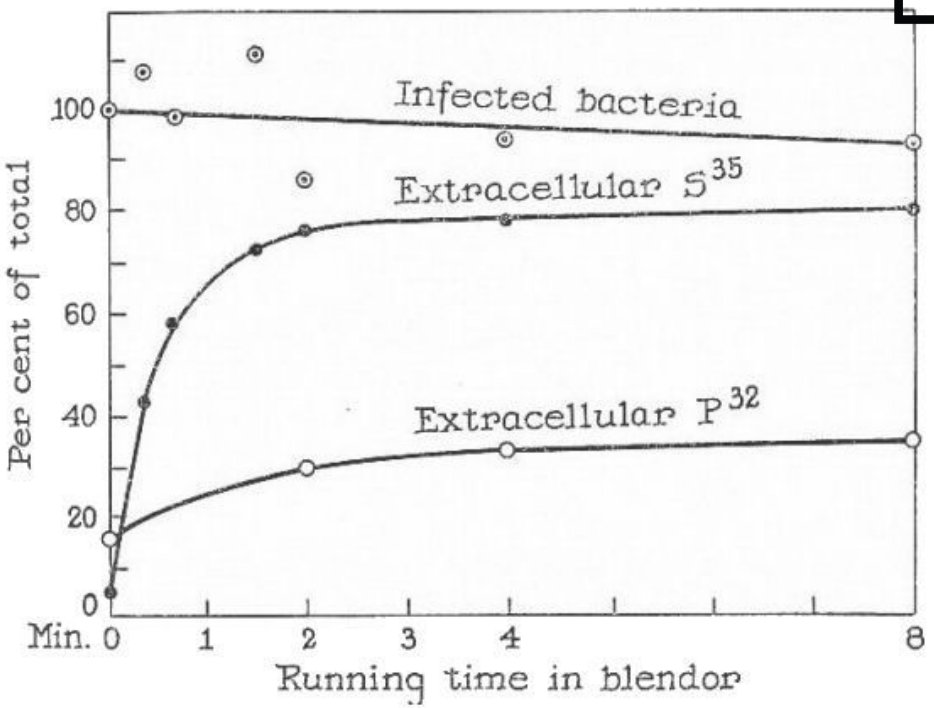
SUPERNATANT VS SEDIMENT



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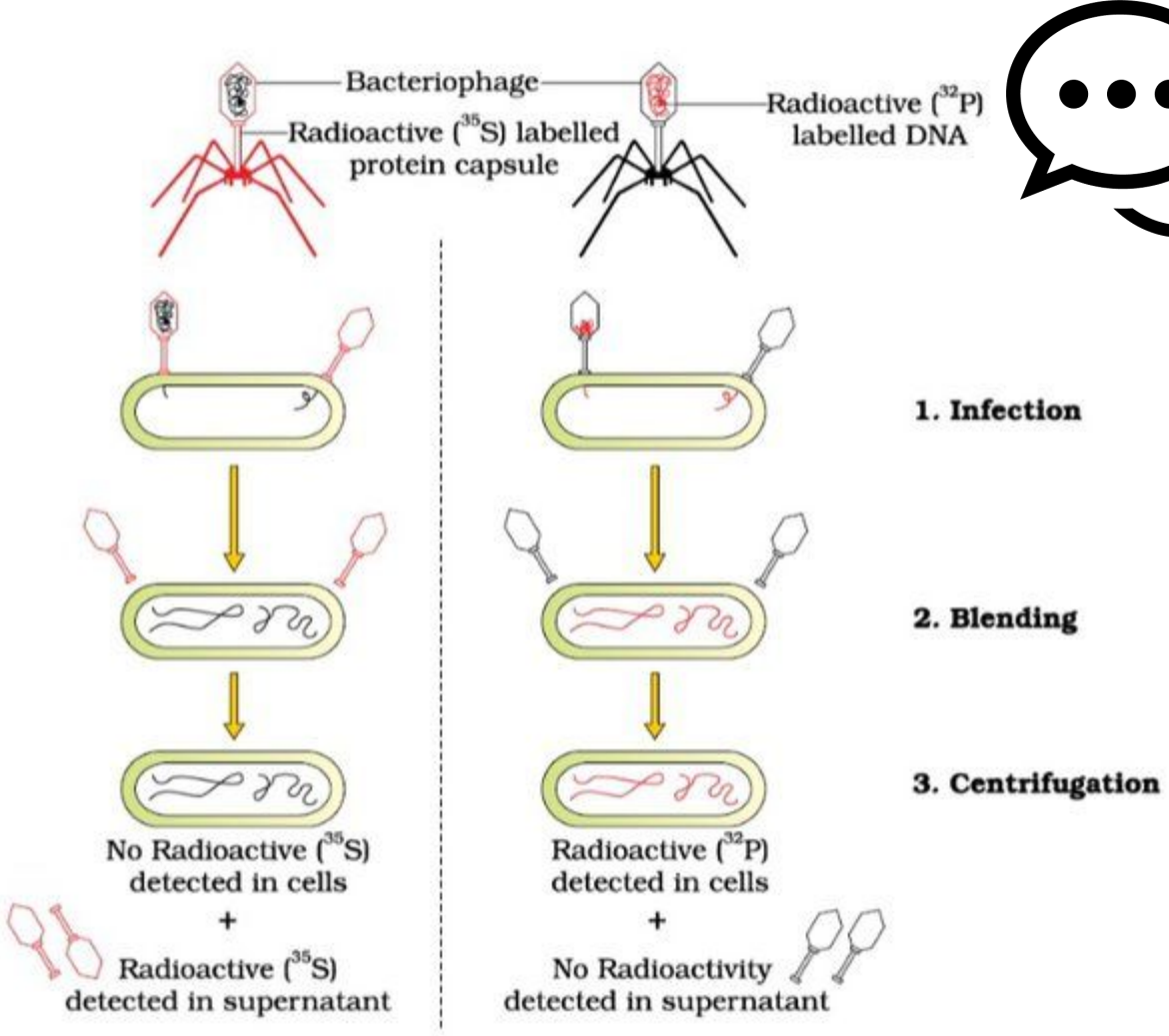
Here extracellular is referring to the isotopes found in the **supernatant**

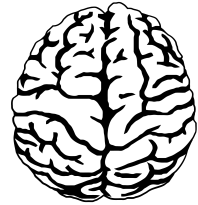
At 4 minutes, how much radioactive phosphate was found inside the cells?



Note: infected bacteria is the number of bacterial cells that survived the blending

Use this diagram to help explain in simple terms the Hershey and Chase experiment





Why could Hershey and Chase not use radioactive nitrogen for their experiment?

Why could Hershey and Chase not use radioactive nitrogen for their experiment?

Both proteins and nucleic acids contain nitrogen!

Theory of Knowledge



To what extent do technological advancements shape the production of knowledge in the natural sciences?

Nature of Science

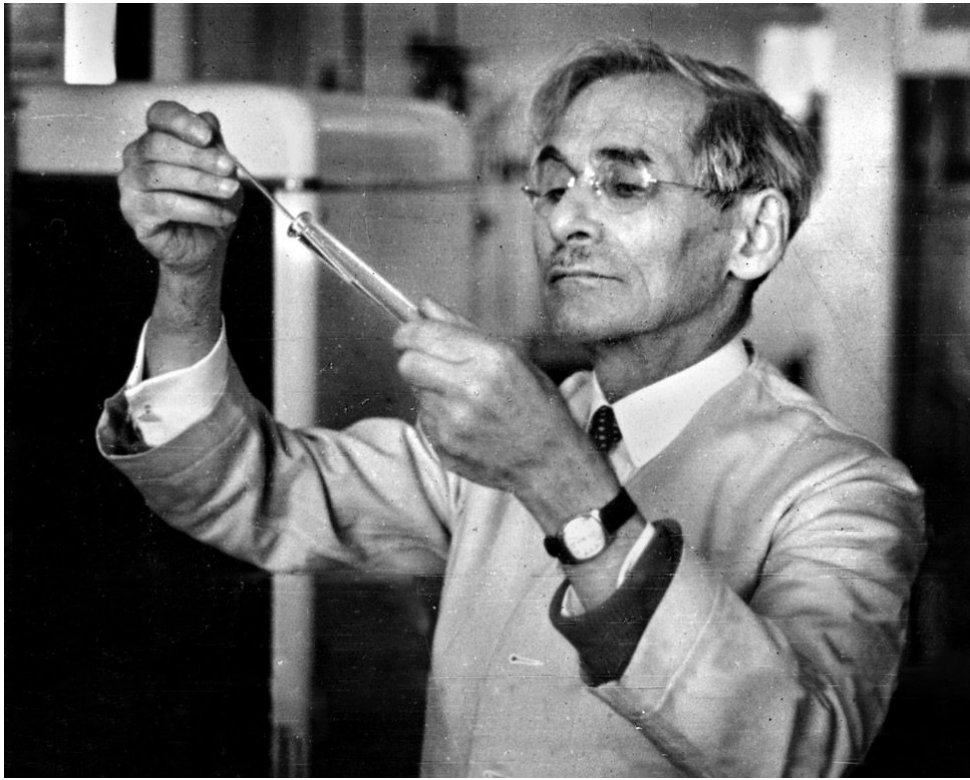
Technological developments can open up new possibilities for experiments. When radioisotopes were made available to scientists as research tools, the Hershey–Chase experiment became possible.

A1.2.15—Chargaff's data on the relative amounts of pyrimidine and purine bases across diverse life forms

Tetranucleotide hypothesis

In 1909 Phoebus Levene proposed that DNA was made up of equal amounts of Adenine, Thymine, Cytosine and Guanine - and that the nucleotides did not carry the genetic information.

Levene saw this pattern and induced that all DNA must follow the same pattern.



Tetranucleotide hypothesis

In 1909 Phoebus Levene proposed that DNA was made up of equal amounts of Adenine, Thymine, Cytosine and Guanine and that the nucleotides were the carriers of genetic information.

Levene saw this pattern and induced that all DNA must follow the same pattern.



Inductive reasoning

means coming to a very broad conclusion based on just a few observations.

The flaw is that no one person can observe all cases of a particular issue.

Inductive reasoning allows you to create a hypothesis to then support or disprove.



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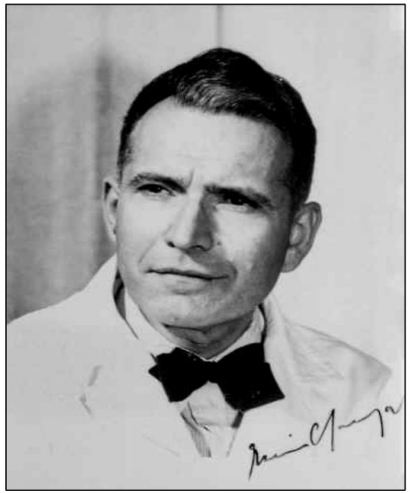


Is inductive reasoning a sufficient basis for forming hypotheses in science?

Chargaff

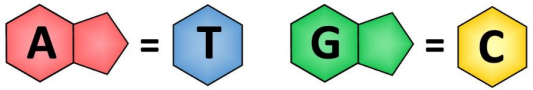
Chargaff's experiments in the 1940s gave crucial information for working out the structure of DNA.

Chargaff found that the relative amounts of A and T, and the relative amounts of C and G were always the same.



	A	T	G	C
Human	31.0	31.5	19.1	18.4
Fruit fly	27.3	27.6	22.6	22.5
Wheat	26.8	27.2	22.8	23.2
Fungus	23.0	23.3	27.1	26.6
Rat	28.7	28.4	21.4	21.5
E. coli	24.6	24.3	25.5	25.6

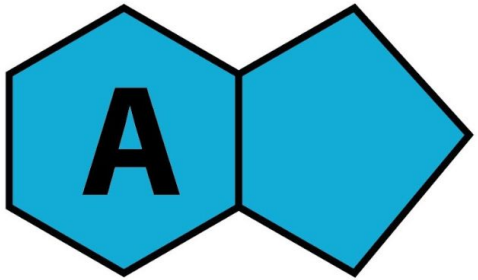
Conclusion: Purines = Pyrimidines



A = T

G = C

A+G = T+C

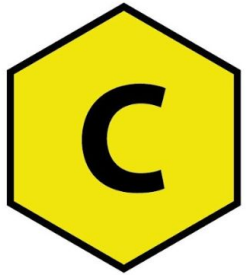


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Purines = **Pyrimidines**

Nature of Science

The “problem of induction” is addressed by the “certainty of falsification”. In this case, Chargaff’s data falsified the tetranucleotide hypothesis that there was a repeating sequence of the four bases in DNA.

A single example of when the proposed theory does not hold true is enough to falsify a hypothesis.



What do you think the percentage of the missing bases would be?

	Adenine	Guanine	Cytosine	Thymine
Human				30%
Chicken	28%			
Rat			21%	
E. coli		27%		

Can you...

Describe the directionality of DNA?

Describe the structure and function of a nucleosome?

Describe the Hershey-Chase experiment?

Describe what Chargaff’s data showed?

